Synthetic Data Generation of linked datasets of the German Socio-Economic Panel

Bachelor thesis

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Ludwig-Maximilians-Universität München
Department of Statistics

Submitted by:

Anar Battaivan

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Supervisor: Dr. Anna-Carolina Haensch

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List of Abbreviations

CART	Classification and Regression Tree
CI	Confidence interval
FCS	Fully conditional specification
IPSO	Information preserving statistical obfuscation
JM	Joint modeling
MAR	Missing at Random
MCAR	Missing completely at random
MICE	Multivariate Imputation by Chained Equations
MNAR	Missing not at random
pMSE	Propensity score mean square error
PRAM	Post randomization method
S_pMSE	Standard propensity score mean square error
SDC	Statistical disclosure control
SOEP	German Socio-Economic Panel
SPECKS	Synthetic data generation; Propensity score matching; Empirical Comparison via the Kolmogorov-Smirnov distance

1 Introduction

Statistical disclosure control (SDC) is an expanding area of research that focuses on protecting sensitive information in data gathered by statistical institutions (Hundepool, 2012). As more data is collected and stored, concerns about confidentiality and privacy rise (Templ, 2017; Templ et al., 2015). Balancing the societal benefits of data access with the potential harms of insufficient data protection is a challenge. Microdata consists of records containing information about individuals, households, and businesses that is utilized in official statistics to produce aggregated data. Privacy laws dictate that such microdata cannot be distributed if it allows for the re-identification of individuals or organizations. Anonymizing microdata is essential to comply with privacy regulations and prevent the assignment of confidential information to specific respondents, especially given the growing need for detailed microdata in research, training, and simulation studies.

Various anonymization approaches exist, one of these is generating fully synthetic data by simulating all variables through statistical models and using predictive distributions. Synthetic datasets contain records with values sampled from probability distributions that preserve relationships, allowing inferences made in the synthetic data to hold true for the original data. Synthetic data generation aims to protect privacy by producing data sets without direct ties to real people or entities while retaining a utility similar to the original data (Drechsler and Haensch, 2023; Templ et al., 2015; Templ, 2017).

This thesis explores the effectiveness of synthetic data generation for complex, longitudinal data sets. Longitudinal or panel data tracks the same sample over different time points. Researchers can better understand the balance between data privacy and utility by assessing how well synthetic data generation works for such data, as it is still an area of ongoing research.

To analyze the effectiveness of synthetic data on longitudinal data, methods for simulating fully synthetic data were used on data provided by the German Socio-Economic Panel (SOEP). As a panel study, the SOEP has been part of the international research infrastructure for almost four decades. It provides annual microdata that tracks biographical developments, which is valuable for basic research and policy advice in the social, economic, and behavioral sciences (Wagner et al., 2008). The SOEP serves as a household panel and a cohort study, enabling researchers to examine changes and trends in various aspects of people's lives over time (Wagner et al., 2008).

1.1 Aim of the Thesis

Anonymization methods, including SDC and synthetic data generation, always involve a tradeoff between minimizing the disclosure risk and maximizing the analytical utility of the resulting dataset. The critical aspect of this thesis is to evaluate the utility and disclosure risk of fully synthetic data generated from the SOEP. The SOEP offers a wide range of datasets, each containing information from different questionnaires on various topics. This thesis focuses on evaluating three datasets derived from personal, household, and health questionnaires. Additionally, a comparison is made to assess the quality of synthetic data when datasets are first synthesized and merged for analysis versus when they are merged before synthesis.

The utility evaluation includes both global utility, which directly compares synthetic data to the original in a distributional sense, and outcome utility, which evaluates usefulness by comparing specific inferential analyses using the synthetic and the original data (Drechsler and Haensch, 2023). To assess the disclosure risk, the concepts of k-anonymity (Sweeney, 2002) and l-diversity (Machanavajjhala et al., 2007) are utilized to evaluate how well the generated synthetic data provide confidentiality and privacy protection in their respective definitions.

1.2 Structure of the Thesis

First, an introduction to the statistical methodology and theoretical background of this thesis is given by covering key topics such as SDC, machine learning, multiple imputation, and synthetic data. It discusses the motivation behind each method and the relevant software tools. The SDC section delves into the classification of variables, various procedures, and concepts like k-anonymity and l-diversity. The following section explores machine learning algorithms, including the classification and regression tree (CART) and linear regression. The multiple imputation section covers missing data mechanisms and procedures such as joint modeling (JM) and fully conditional specification (FCS). The synthetic data section focuses on motivation, procedure, utility evaluation (global and outcome), and software tools. The next chapter gives an introduction to the German Socio-Economic Panel (SOEP) and its relevance to the study. Following that, the results chapter showcases the findings and analysis in this thesis. Finally, the last chapter concludes the paper by summarizing the main points and implications.

2 Statistical Methodology and Theoretical Background

This chapter gives an overview of the used statistical methods and reviews the necessary background knowledge. Since the analysis part of this thesis was performed on panel data, the priority in explaining the motivational aspects and benefits of the methodology is more focused on the context of survey statistics.

2.1 Statistical disclosure control (SDC)

Although the demand for sensitive information from sources like surveys, censuses, and registers has grown significantly in recent times, it is crucial to maintain the privacy and confidentiality of information of individual units before sharing data with the public or researchers. This can be done by applying SDC methods to minimize the potential disclosure risk associated with the data (Templ, 2017). However, implementing SDC methods on the original microdata leads to a loss of information, that impacts the data's utility, affecting the data's value for analysis in completeness and accuracy. Thus, statistical agencies are confronted with the task of finding the best SDC methods to minimize the risk of disclosure while causing the least amount of information loss, thereby maintaining the data's utility (Templ, 2017).

2.1.1 Classification of Variables

Templ (2017, Section 2.1) describes that various types of variables need to be distinguished in the context of SDC.

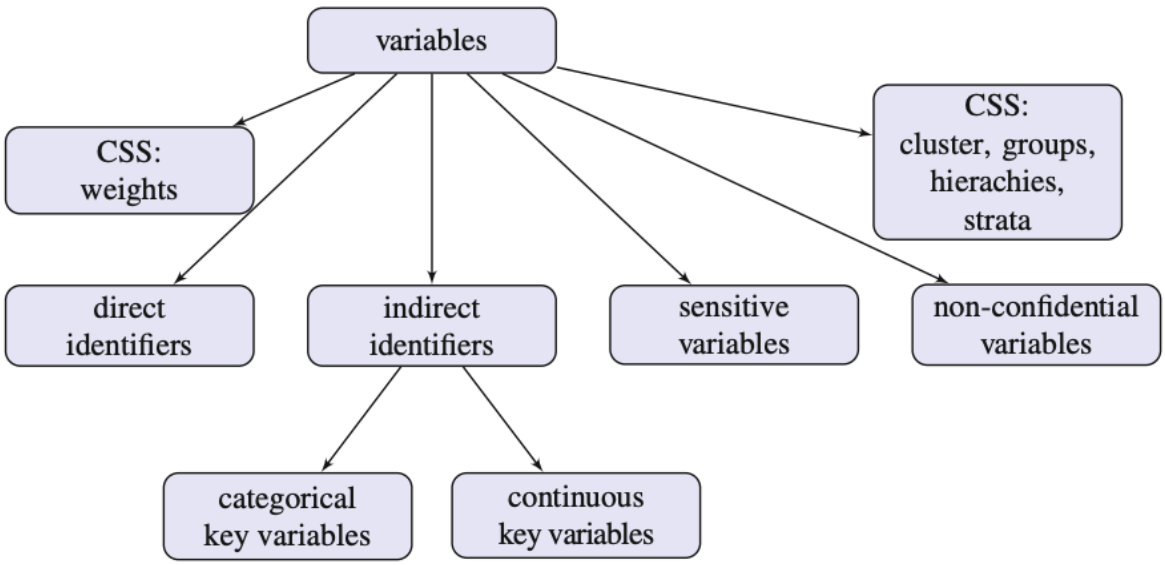


Figure 1: Types of variables (Templ, 2017, Page 36 Fig. 2.1)

- Non-confidential variables: These variables cannot be linked to external information sources, nor is any data available in external databases (Templ, 2017).
- Identifying variables: Identifying variables can be categorized into two types: direct identifiers and key variables (Templ, 2017).

  Direct identifiers can distinctly recognize statistical units, for example social security numbers or names of individuals. The initial step of SDC involves removing these direct identifiers (Templ, 2017).

  Key variables / indirect identifiers in combined form can be linked to external data to identify individuals within the published dataset (Hundepool, 2012; Templ, 2017). Although variables such as gender, age, region, and occupation do not allow for individual identification on their own, their combination may enable the unique identification of individuals (Templ, 2017).
- Sensitive variables: For these variables, it is crucial to ensure that the information cannot be linked back to any specific individual. These variables can be sensitive and might also serve as direct identifiers or key variables. Such sensitive variables include details about an individual's criminal history, sexual behavior, medical data, or income (Templ, 2017).
- Sampling weights: Sampling weights represent the probability of each person being selected to participate in the survey. When dealing with a sample rather than the entire population, there is an additional layer of uncertainty, as it is generally unknown if a specific person is part of the survey sample. If a unique unit in the population is represented in the sample, it may have the same risk as if the entire population was sampled (Templ, 2017).
- Hierarchies and clusters: Hierarchies and clusters may appear in data depending on the sampling design and need to be considered in SDC methods since the estimation of disclosure risk is different when such clusters are present compared to when they are not. (Templ, 2017).

2.1.2 Procedure

Following the workflow presented by Templ et al. (2015, Section 2.3) and illustrated in Figure 2, the general procedure of creating confidential microdata can be summarized in following steps:

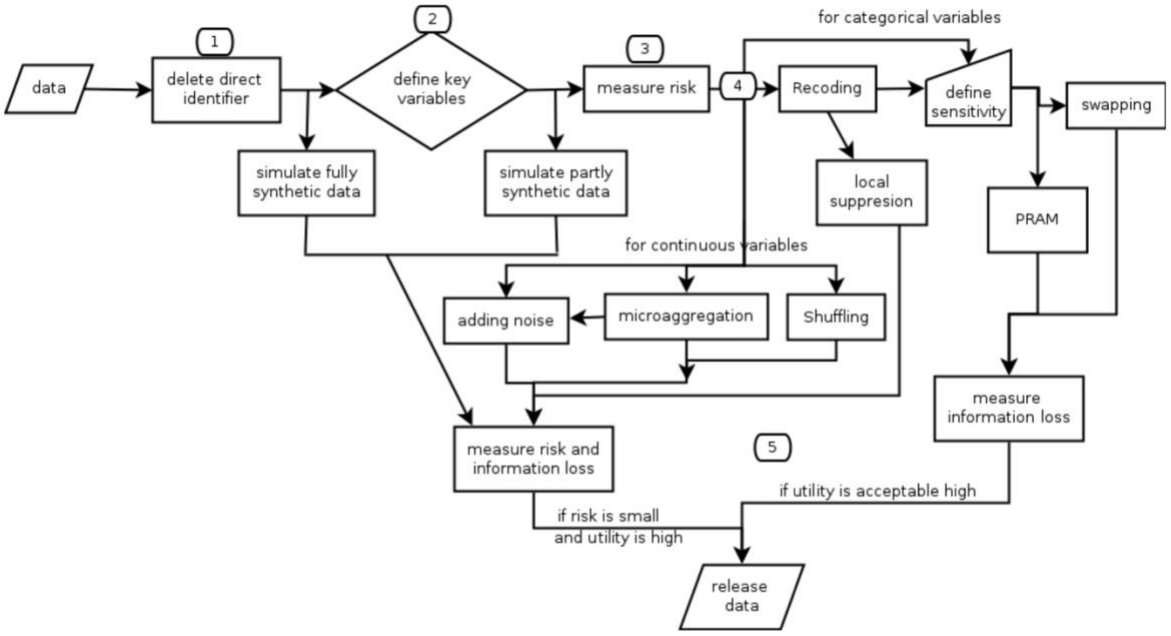


Figure 2: Workflow to anonymize microdata using SDC methods (Templ et al., 2015, Page 5 Figure 1)

1. The initial step consists of removing all direct identification variables (Templ et al., 2015).

2. The second step involves selecting a set of key variables for further risk calculations. This decision is not objective and requires discussions with experts on the subject matter and an understanding of relevant national laws. The selection of key variables is essential to ensure accurate risk assessment and compliance with legal requirements (Templ et al., 2015). It should also be noted that choosing key variables is unnecessary for creating fully synthetic data, as seen in Figure 1. See Section 2.4 for further information on this method.

3. After selecting key variables, the disclosure risks of individual units are measured using sample frequency counts and applying probability methods, which account for population frequencies, to estimate re-identification risks (Templ et al., 2015).

4. Recoding, local suppression, swapping, and post-randomization (PRAM) can be applied to categorical key variables to address high-risk observations. The choice depends on the structure of key variables. For continuous key variables, perturbation is needed, with micro aggregation or shuffling as potential methods. While these methods are not used in this thesis, further information can be found in Templ et al. (2015).

5. Information loss and disclosure risk are assessed once the key variables are modified. The goal is to release a secure dataset with low risks and high utility. If the individual risks are sufficiently low, and the utility is high enough, the data set can be released; otherwise, the process is repeated with additional perturbations to improve security and utility (Templ et al., 2015).

2.1.3 K-Anonymity

Unique observations in a dataset must be better protected to protect confidentiality as they have a high risk of identification. K-anonymity is a privacy protection approach that ensures that each unique combination of key variables in a dataset is shared by at least *k* records in the micro-dataset (Templ, 2017). By guaranteeing that each unique pattern of key variables is present in at least k records, it becomes more difficult to single out and identify any specific individual in the dataset, thus protecting confidentiality and privacy (Templ, 2017).

2.1.4 L-Diversity

L-diversity is an extension of k-anonymity that considers the distribution of sensitive variables among a set of records sharing the same pattern in key variables (Templ, 2017). While k-anonymity prevents the identification of an individual within a group, it does not account for situations where each record in a set shares an identical value for a sensitive variable (e.g., cancer in the medical diagnosis variable) (Raghunathan et al., 2017). In such cases, if an attacker can identify one individual in the group, they will know that person's sensitive information with certainty. L-diversity addresses this issue by ensuring a diverse distribution of the sensitive variable within the group (Templ et al., 2015).

2.1.5 Statistical Software

To analyze the disclosure risk of the synthetic data via k-anonymity and l-diversity in Chapter 4, the R package sdcMicro[1] version 5.7.5 is used. SdcMicro is a package that allows for the generation of anonymized microdata via various SDC methods implemented into functions in the package (Templ et al., 2015). The *createSdcObj()* function creates an object of class sdcMicroObj by defining parameters such as categorical and continuous key variables, sensitive variables, and more from the microdata. Applying *ldiversity(),* the functions return the l-diversity measures for the given object of class sdcMicroObj.

[1] See *CRAN - Package sdcMicro*. Retrieved from https://cran.r-project.org/package=sdcMicro for a more detailed explanation of the package.

2.2 Machine Learning

Machine learning uses statistical models to process vast data volumes and identify patterns and relationships for predictive purposes. Machine learning algorithms can be categorized based on their feedback type, including supervised, unsupervised, and reinforcement learning. Lang and Zhong (2022) describe these three categorizations as follows:

Supervised learning is machine learning, where algorithms are based on data with standard inputs and outputs. The use of the data of known labels and classes is supervised (Lang and Zhong, 2022).

On the other hand, unsupervised learning algorithms work with data that only contains inputs. There are no known or expected outputs. These algorithms create mathematical models to discover hidden structures, patterns, and relationships within the data (Lang and Zhong, 2022).

Reinforcement learning is a different machine learning approach where an algorithm learns to make decisions by trial and error, taking into account the feedback it receives from its actions. It aims to maximize rewards based on predefined rewards and punishments, improving its problem-solving strategies (Lang and Zhong, 2022).

2.2.1 Algorithms

Two supervised learning algorithms are further explained in this section, as they are used in the analyses in Chapter 4.

2.2.1.1 Classification and Regression Tree (CART)

The CART algorithm is a decision tree learning approach that can be used with any data (Nowok, 2016). As fully explained by Nowok (2016), the main idea is to repeatedly divide a dataset into smaller groups based on specific conditions, creating increasingly homogeneous subsets regarding the outcome variable. These splits can be visualized as a tree structure, with each split acting like a yes-or-no question about one or more predictor variables. The goal is to create groups (leaves of the tree) where the conditional distribution of the outcome variable is relatively homogeneous. This means that, given the specific conditions that define a group, the outcome variable behaves similarly for all units in that group. When creating synthetic data, the values are generated by sampling from the appropriate group. The original CART algorithm decides on the best split at each step by testing the null hypothesis of independence between the outcome variable and all explanatory variables (Nowok, 2016). If this hypothesis cannot be rejected, it stops further splitting. If rejected, it selects the explanatory variable with the strongest association with the

outcome variable to do the split. This process is recursively repeated for each subgroup (Nowok, 2016).

2.2.1.2 Linear Regression

Linear regression is a commonly used method in statistics and machine learning for predicting an outcome variable (Y) based on one or more predictor variables (X), assuming a linear relationship is given between them. It is referred to as a single regression when there is only one input variable. If there are multiple input variables, it is a multivariate multiple regression. The model is represented by the following equation (Küchenhoff, 2021):

$$Y = \beta_0 + \beta_1 X_1 + ß_2 X2 + \cdots + \beta_p X_p + \epsilon$$

Here, β represents the coefficients for each input variable estimated during the model's training. β_0 is an additional coefficient known as the intercept. The ε term represents the error or noise in the model, following a normal distribution by assumption with the mean $\mu = 0$ and some standard deviation σ.

2.3 Multiple Imputation

Multiple imputation is extensively utilized across diverse fields and sectors, such as healthcare, finance, social sciences, and more. By aiding researchers, analysts, and data scientists in handling incomplete datasets, it paves the way for improved decision-making and efficient solutions to address real-world challenges. The purpose of this section is to give an overview of the concept behind multiple imputation. It first discusses the motivational aspect of why the use of multiple imputation would prove beneficial in survey statistics; it then explains the concepts of missing data mechanisms, and the procedure of multiple imputation, and it finally gives an introduction to the used statistical software package.

2.3.1 Motivation

The occurrence of measurement errors in surveys has become a persistent and unavoidable issue in empirical social science. While measurement errors can be caused by various factors, such as incorrect responses from participants, poorly designed questionnaires, or even technical malfunctions, these results vary in inaccurate and incomplete data (Böwing-Schmalenbrock and Jurczok, 2011). Although it is possible to reduce the frequency and magnitude of these errors through careful planning and execution of the surveys, it remains impossible to eliminate them. Neglecting inconsistent or missing data can lead to a loss of valuable information and weaken the validity of data analyses (Böwing-Schmalenbrock and Jurczok, 2011).

While there are other approaches to adjust for missing data, for example complete case analysis (listwise deletion) and single imputation, multiple imputation has proven to have the best statistical properties (StataCorp, 2021). A complete case analysis that ignores missing data and only takes survey participants with complete data into account has the benefit of being convenient, but it runs into the problem of precision loss as the sample size becomes reduced while also potentially running into the dangers of biased results and systematic errors (Böwing-Schmalenbrock and Jurczok, 2011; Buuren, 2018). In contrast, imputation deals with missing values by substituting missing values with fair values based on the used method and model. Single imputation, which represents a single set of plausible values for an imputation step, has the flaw of treating the missing values as known in the analysis, leading to an overly optimistic confidence interval and significance test results due to an underestimation of the variances of the estimates (Buuren, 2018; StataCorp, 2021). However, multiple imputation, which represents multiple sets of plausible values performed at the imputation step, balances this issue out as it considers the uncertainty and sampling variability due to missing data (Buuren, 2018; Vink, 2022).

In summary, multiple imputation comes with great benefits. The main benefits are the preseveration of the sample size and the ability to account for uncertainty in missing values, leading to better and more accurate estimators, and allowing for valid statistical inferences, including confidence intervals and p-values.

2.3.2 Missing data mechanisms

An essential aspect of missing data is the missing data mechanism. Rubin (1976) introduced three primary types of missingness mechanisms: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR).

Adopting the notations introduced by Buuren (2018), let R be a binary response indicator matrix that stores the locations of missing data in $Y := (Y_{obs}, Y_{mis})$ and let ψ contain the parameters of the missing data model $\Pr(R|Y_{obs}, Y_{mis}, \psi)$. Y_{obs} collectively represents the observed data; Y_{mis} collectively represents the missing data, and Y represents the complete data. Y and R are both matrices of the dimension $n \times p$ denoted by their elements y_{ij} and r_{ij} respectively with $i = 1, \dots, n$ and $j = 1, \dots, p$. If $y_{ij} \in Y_{obs}$, then $r_{ij} = 1$, otherwise if $y_{ij} \in Y_{mis}$, then $r_{ij} = 0$. The missing data model describes the relationship between the distribution of R and the location of the missing data Y (Buuren, 2018).

MCAR means that the probability of missingness is equal in all cases, making it unrelated to the observed and unobserved data (Buuren, 2018).

$$\Pr(R = 0|Y_{obs}, Y_{mis}, \psi) = \Pr(R = 0|\psi)$$

MAR is based on the probability of the missingness being equal only within groups defined by the observed data, making it dependent on the observed but not the unobserved data (Buuren, 2018).

$$\Pr(R = 0|Y_{obs}, Y_{mis}, \psi) = \Pr(R = 0|Y_{obs}, \psi)$$

MNAR has varying probabilities as it depends on the missing data itself. Therefore, the missing data model $\Pr(R = 0|Y_{obs}, Y_{mis}, \psi)$ cannot be further simplified (Buuren, 2018).

While MCAR makes it possible to ignore many complexities that arise when handling missing data and allows more straightforward methods to work, it is quite restrictive and unrealistic to be encountered. Multiple imputation can handle both the MAR and MNAR missing data mechanism. However, MNAR is the most complex case of missing data mechanism, whereas many modern methods for dealing with missing data rely on the MAR assumption (Buuren, 2018). Under the condition of the missing data being MAR and the parameters of both the data model and the missing data mechanism being distinct, the missing data mechanism is said to be ignorable (Buuren, 2018; StataCorp, 2021). Thanks to this assumption, the process causing missing data in the imputation model can be ignored, leading to a simplification in the imputation step while preserving the correctness of the inference (StataCorp, 2021). On the other hand, MNAR requires additional modeling assumptions in the multiple imputation, influencing the generated imputations. To deal with MNAR missing data, one could either expand the data in the imputation model, hoping it gets closer to being MAR, or perform a sensitivity analysis on the critical parameters of a nonignorable imputation model, an analysis repeated under various missing data models (Buuren, 2018; Buuren and Groothuis-Oudshoorn, 2011). Testing the MAR missing data mechanism against the MCAR missing data mechanism is impossible as the information needed to distinguish between these two types can only be obtained through the unobserved (Buuren, 2018; StataCorp, 2021).

The assumption for the missing data inside the SOEP is MAR for all following chapters. Since many methods of synthetic data generation rely on imputation methods that require a MAR or MCAR assumption, the analyses on the panel in Chapter 4 can proceed with this assumption.

2.3.3 Procedure

The analysis begins with incomplete data containing the observed data. Multiple imputation substitutes the missing values in a dataset by m > 1 simulated versions produced in alignment with a probability distribution $\Pr(Y_{mis}|Y_{obs})$ for missing values given the observed values (Drechsler,

2010). Each of the m imputed datasets is then analyzed individually by standard methods designed for complete data (Drechsler, 2010). Ultimately, the results of all m evaluations are pooled, thus consolidating the m estimates, confidence intervals, and test statistics into a single result, that properly represents the imputation's inherent uncertainty (Buuren, 2018; Drechsler, 2011). Figure 1 illustrates the multiple imputation steps.

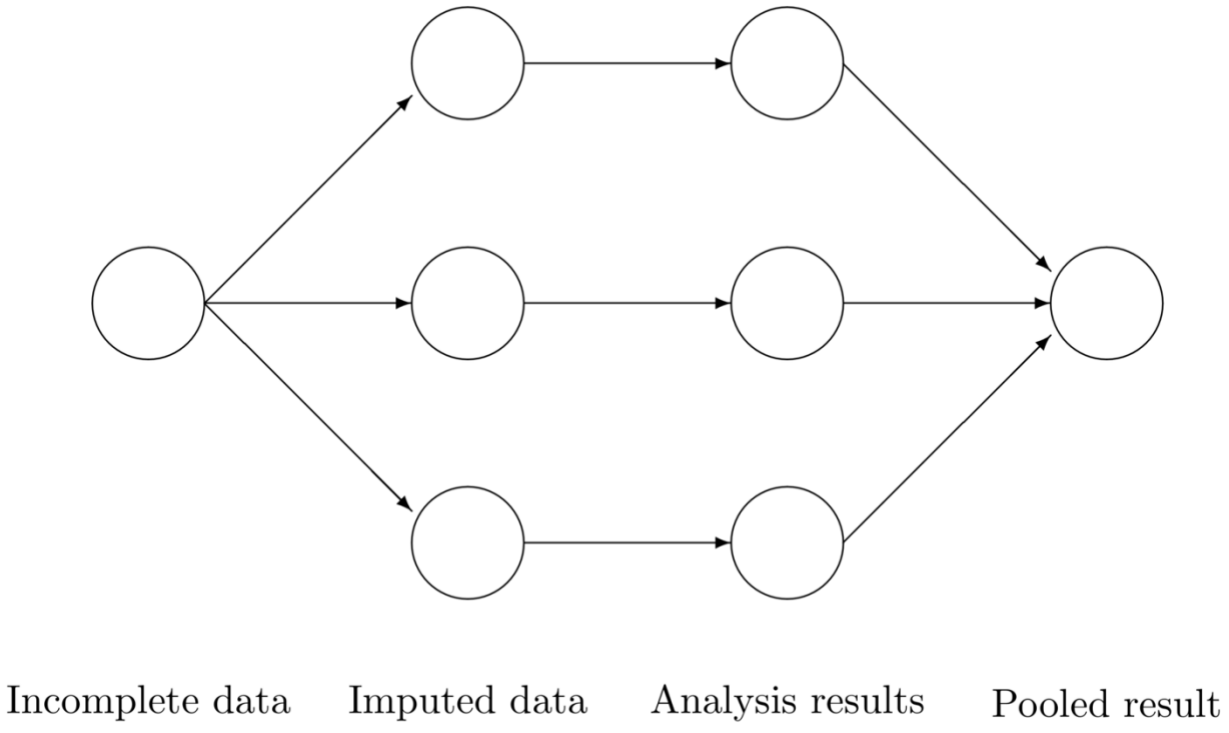


Figure 3: Schematic illustration of the main steps in multiple imputation with m = 3 by Buuren (2018, Chapter 1.4)

While there are various approaches to multiple imputation with random missing data patterns, the primary methods are joint modeling and fully conditional specification (StataCorp, 2021).

2.3.3.1 Joint modeling (JM)

JM assumes an authentic multivariate distribution for all imputation variables (Drechsler, 2011). Given a two-step approach, it first substitutes the parameters from the underlying distribution, represented by θ, by random draws derived from the subsequent posterior distribution $\Pr(\theta|Y_{obs})$ given the observed values Y_{obs}. Afterward, it substitutes the missing values Y_{mis} from the conditional predictive distribution $\Pr(Y_{mis}|Y_{obs},\theta)$ given the parameter θ derived in the previous step (Drechsler, 2011; Schafer, 1997).

$$\Pr(Y_{mis}|Y_{obs}) = \int \Pr(Y_{mis},\theta|Y_{obs})\,d\theta = \int \Pr(Y_{mis}|Y_{obs},\theta)\Pr(\theta|Y_{obs})\,d\theta$$

2.3.3.2 Fully conditional specification (FCS)

FCS refrains from assuming multivariate distributions for all imputation variables (StataCorp, 2021). It rather offers a set of chained equations, which are univariate conditional distributions Pr $(Y_j|Y_{-j}, \theta_j)$ of each variable with fully conditional descriptions of predictive equations (StataCorp, 2021). Y_j is the j^{th}column in Y and Y_{-j} is the complement of Y_j (Drechsler, 2011, Chapter 3).

Rubin (2004) broke down the effort required to generate imputations into three components:

- Modeling component: Select a specific data model.
- Estimation component: Establish the posterior parameter distribution based on the model.
- Imputation component: Take random draws for the missing data by successively drawing from parameter and data distributions.

By directly defining the conditional distributions from which samples should be obtained, FCS circumvents the need to specify a multivariate model for the data.

2.3.4 Statistical Software

The analyses in Chapter 4 will work with the MICE package[2] version 3.15.0 in R, which implements a method to create multiple imputations for multivariate missing values based on the FCS approach (Buuren and Groothuis-Oudshoorn, 2011). Let X denote all explanatory variables for the imputation and let $\phi = \phi_1, \dots, \phi_p$ be a parameter space, where ϕ_j represents the unknown parameters of the jth imputation model. Buuren (2018, Chapter 4.5.2, Algorithm 4.3) then explains the multivariate imputation by chained equations (MICE) algorithm as follows:

> ***"Algorithm 4.3 (MICE algorithm for imputation of multivariate data)***
>
> 1. *Specify an imputation model* $Pr\left(Y_j^{mis} \middle| Y_j^{obs}, Y_{-j}, R\right)$ *for variable* Y_j *with* $j = 1, \dots, p$.
> 2. *For each* j*, fill in starting imputations* $\dot{Y}_j^0$ *by random draws from* Y_j^{obs}.
> 3. *Repeat for* $t = 1, \dots, M$.
> 4. *Repeat for* $j = 1, \dots, p$.

[2] See *CRAN - Package MICE*. Retrieved from https://cran.r-project.org/package=mice for a more detailed explanation of the package.

5. *Define* $\dot{Y}_{-j}^{t} = (\dot{Y}_{1}^{t}, \ldots, \dot{Y}_{j-1}^{t}, \dot{Y}_{j+1}^{t-1}, \ldots, \dot{Y}_{p}^{t-1})$.
6. *Draw* $\dot{\phi}_{j}^{t} \sim Pr\ (\phi_{j}^{t} | Y_{j}^{obs}, \dot{Y}_{-j}^{t}, R)$.
7. *Draw imputations* $\dot{Y}_{j}^{t} \sim Pr\ (Y_{j}^{mis} | Y_{j}^{obs}, \dot{Y}_{-j}^{t}, R, \dot{\phi}_{j}^{t})$.
8. *End repeat j.*
9. *End repeat t.“*

The package implements the MICE algorithm and saves the outcomes of each imputation step within distinct, specific data classes: mids, mira, and mipo. Figure 4 illustrates the procedure explained in Section 2.3.3 in the context of different functions native to the MICE package. Beginning with an incomplete dataset stored as a data frame, the *mice()* function creates m > 1 multiple imputed datasets stored as the special class mids in the package. The multiple imputed datasets are then individually analyzed via chosen inference methods and models designed for complete data via the *with()* function. The multiple imputed repeated analysis results are stored as an object of class mira. In the last step, the *pool()* function combines the analysis results using methods outlined in Rubin's (2004) rules and stores the multiply imputed pooled outcomes as an object of class mipo (Buuren and Groothuis-Oudshoorn, 2011).

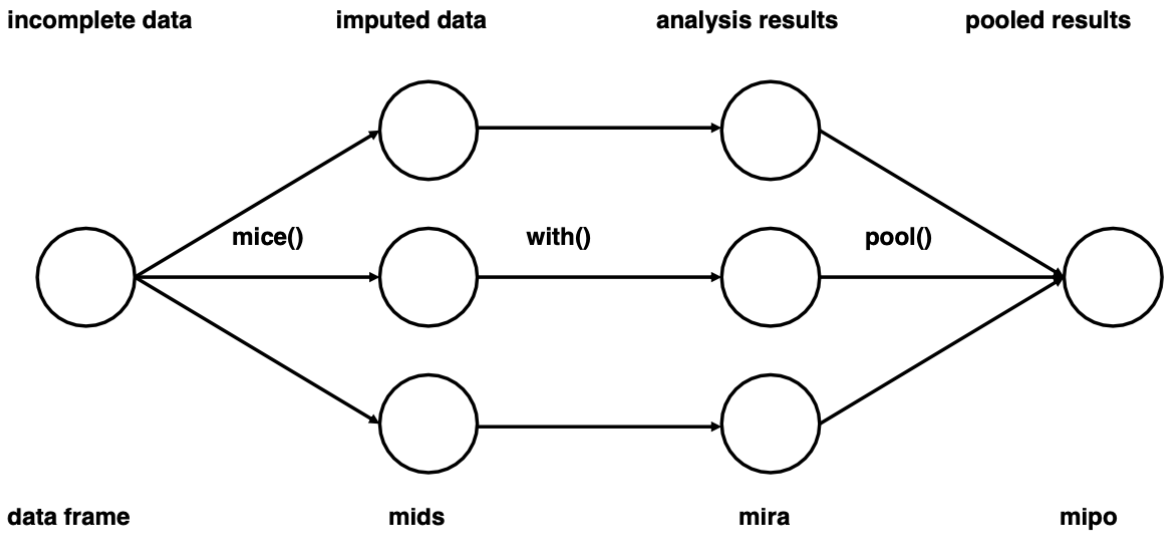


Figure 4: Schematic illustration of the multiple imputation steps performed by the MICE package via the functions mice(), with() and pool() showing the resulting data storage types for each step. In this example, m = 3 (Buuren and Groothuis-Oudshoorn, 2011, Page5).

2.4 Synthetic Data

As concerns about data privacy and regulatory compliance grow, synthetic data offers a solution for creating datasets that closely mirror real-world data while safeguarding personal privacy. This section introduces the topic of synthetic data by discussing the motivational factors behind developing and using this method, explaining its procedure, and giving an exemplary approach to implementing the processes in a statistical software environment.

2.4.1 Motivation

The idea of fully synthetic data for privacy protection was first introduced by Rubin (1993). Building upon his previous research in multiple imputation for U.S. Census Bureau projects, he proposed a method to create synthetic data by treating unselected population units as missing data, employing multiple imputation techniques, and releasing random samples from these imputed populations to the public with a significantly reduced disclosure risk (Drechsler, 2011). At the same time, most of the valuable information from the original data is preserved, provided that the imputation models used in the process are chosen with care and precision. However, since the quality of fully synthetic data relies heavily on the quality of the model used to generate it, identifying a model that accurately represents the complex relationships inside data consisting of a vast number of variables with nested logical constraints between them is a difficult task (Drechsler, 2011).

Little (1993) proposed a partially synthetic data method similar to Rubin's. The critical difference is that only specific parts of the data are replaced, for example sensitive variables or key variables that could identify a single entry in the dataset. By selectively protecting sensitive records and leaving other values unchanged, the overall quality of the data is maintained. However, this approach comes with the downside that the unaltered values in the dataset carry a higher risk of disclosing private or sensitive information (Drechsler, 2011).

2.4.2 Procedure

Methods based on the JM multiple imputation method focus on directly defining the data's joint distribution. Early methods like information preserving statistical obfuscation (IPSO) (Burridge, 2003) were based on the assumption of multivariate normality, which is often not applicable to real-world data. However, more recent methods have introduced flexible models that better address the complexities of actual data, such as a method based on the Dirichlet process mixture

of products of multinomials (DPMPM) (Drechsler and Haensch, 2023; Hu et al., 2014) for categorical data.

Based upon the FCS method of multiple imputation, each variable is synthesized one at a time, the models taking into account any previously synthesized variables or those remaining unchanged in the final dataset. The core concept behind this approach is that a joint distribution can be expressed as a series of conditional distributions, allowing for the step-by-step synthesis of variables (Drechsler and Haensch, 2023). This also means that the sequential regression approach is highly adaptable because it will enable the use of various models for each variable. Examples include parametric models like linear regression, logit models, and quantile regression models (Drechsler and Haensch, 2023). Furthermore, machine learning tools like CART or random forests, which can generate random draws from conditional distributions, can also be employed in this approach, contributing to its flexibility (Drechsler and Haensch, 2023).

2.4.3 Utility Evaluation

Utility evaluation in synthetic data refers to examining how effectively the synthetic data maintains the original data's statistical characteristics and informational content. This evaluation is critical for determining whether the synthetic data can be effectively used for analysis and modeling, ensuring that it serves its purpose for researchers and analysts by allowing valid conclusions and insights from the synthetic data while protecting sensitive information.

2.4.3.1 Global Utility

The global utility measures the usefulness of synthetic data by comparing it with the original data. One typical approach is the use of distance measures (Drechsler and Haensch, 2023). However, calculating these measures can be challenging for extensive datasets, leading to exploring alternate strategies. Another method examines the difficulty of distinguishing between original and synthetic data by drawing inspiration from propensity score matching (Raab et al., 2021). This technique involves calculating propensity scores by merging original and synthetic records, each marked with an indicator to identify whether it is synthetic, and then fitting a model to estimate the likelihood of each record being part of the synthetic data (i.e., the propensity score). If the synthetic data perfectly replicates the original data, the propensity score distributions for both datasets should be identical (Drechsler and Haensch, 2023).

Various metrics can be used to analyze the differences in propensity score distributions between original and synthetic data (Drechsler and Haensch, 2023). In this thesis, the focus lies on three

prominent evaluation metrics: the propensity score mean square error (pMSE), the standard propensity score mean square error (S_pMSE), and the Kolmogorov-Smirnov distance (SPECKS).

Let p_i be the propensity score of record $i = 1, \dots, N$ in the combined dataset of size $N = n_{org} + n_{syn}$. The pMSE is calculated as the averaged squared differences between the propensity scores of each corresponding observation $\frac{1}{N}\sum_N (p_i - c)^2$, where $c = \frac{n_{syn}}{N}$ (Drechsler and Haensch, 2023). If the synthesis uses a model that matches the original data distribution, the expected mean of p_i will be c. Hence the smaller the pMSE, the closer the distribution and, therefore, the higher the analytical validity of the synthetic data (Drechsler and Haensch, 2023; Raab et al., 2021).

However, the pMSE tends to increase as more predictors are added to the model, which might prove to be a problem regarding the comparability of this metric (Raab et al., 2021). To address this issue, S_pMSE is calculated as a ratio by subtracting the expected value of the pMSE under the null hypothesis from the actual pMSE and then dividing the result by the standard deviation under the null hypothesis (Drechsler and Haensch, 2023). The null hypothesis, in this case, assumes that the synthesis model has been correctly specified.

The SPECKS metric evaluates the difference between the propensity score distributions of the original and synthetic data by calculating the maximum difference between their cumulative distribution functions (Raab et al., 2021). From 0 to 1, lower values indicate better preservation of the characteristics of the original data.

2.4.3.2 Outcome Utility

The outcome utility measures the usefulness of synthetic data for specific tasks. Drechsler and Haensch (2023) present various measures concerning the evaluation of the outcome utility. Analytical validity can be visualized by plotting estimates of interest from both original and synthetic data, with high utility showing a clustering around the 45-degree line (Drechsler and Haensch, 2023). Although this runs into the downside of not accounting for the uncertainty of the estimates.

The confidence interval overlap measure (Drechsler and Haensch, 2023) considers the uncertainty of the estimates and calculates the relative average overlap between the confidence intervals from both datasets. A confidence interval overlap close to 1 indicates similar inferential conclusions from both datasets (Drechsler and Haensch, 2023).

Machine learning efficacy evaluates the similarity of results produced by machine learning models trained on synthetic data compared those trained on original data (Drechsler and Haensch, 2023).

The high utility is indicated by similar evaluation criteria regardless of the training data and sometimes based on whether the synthetic data leads to the same ranking of machine learning models as the original data (Drechsler and Haensch, 2023).

2.4.4 Statistical Software

The tool used in this thesis to generate synthetic versions of microdata is the synthpop package[3] version 1.8.0 for R, which takes the foundation provided by the MICE package (see Section 2.3.4) and enhances it for the specific needs of synthetic data generation and analysis (Nowok et al., 2016). The *syn()* function creates a synthetic dataset by reading in a data frame or matrix to be synthesized. It produces one dataset without sampling from the posterior distribution of the model parameters (Nowok et al., 2016). The m parameter can be adjusted to the desired number of data sets for generating multiple datasets. The global utility can be evaluated using the *utility.gen()* function.

[3] See *CRAN - Package synthpop*. Retrieved from https://cran.r-project.org/package=synthpop for a more detailed explanation of the package.

3 The German Socio-Economic Panel (SOEP)

In his economic sciences nobel lecture in 2015, Angus Deaton highlighted the importance of measuring differences in resources across individuals over their lifetime and linking economic measures of well-being with other social science measures (Deaton, 2016). The German Socio-Economic Panel (SOEP) aligns with these ideas, providing longitudinal data of private households in Germany since 1984 (Goebel et al., 2019). With data from nearly 15,000 households and 30,000 individuals, SOEP covers a wide range of objective and subjective variables, enabling researchers to explore long-term societal changes, intergenerational mobility, and the effects of policy reforms. SOEP is also unique in capturing German unification, making it an invaluable resource for understanding human behavior and decision making in various social and institutional contexts (Goebel et al., 2019).

The SOEP dataset can be classified into various content classifications, including tracking, original, survey, generated, and spell data. Tracking and survey data files outline the development of the sample, showing which individuals or households participated in each year. Original data files contain yearly questionnaire data with minimal consistency checks, while generated data files offer consistently coded variables across all waves for easier use. The dataset also includes biographical data, which can be unchanging or subject to updates. Updated data might be stored as spell data, including details on spell type, beginning and end points, and censoring status.

4 Results

4.1 Data and fully synthetic data generation

To simulate fully synthetic datasets from the information derived from the SOEP, three generated datasets have been selected for the simulation process: pgen, hgen, and pequiv (Grabka, 2020; SOEP Group, 2021, 2021). Generated datasets receive particular attention from the SOEP team, as they are subject to consistency checks and harmonization to improve data quality. They also contain newly generated variables of particular research importance. The pgen dataset contains data on a personal level combined from numerous sources (SOEP Group, 2021), while the hgen dataset offers data from the SOEP household questionnaire (SOEP Group, 2021). The pequiv dataset contains variables harmonized for international comparisons with several panel studies from other countries (Grabka, 2020). As for the data preparation steps, while these datasets contain information from all waves from as early as 1984 up until the year 2021, the data has been reduced to only its first wave due to the vast volume of the data to allow for a more time efficient execution of analysis for this thesis. Removing variables with only missing value entries due to the information not being part of the first wave of questionnaires, we are left with sample sizes of 12245 for the pgen dataset with 48 variables, 5921 for the hgen dataset with 35 variables, and 16252 for the pequiv dataset with 199 variables. By aggregating the hgen and pequiv datasets into the pgen dataset using the personal identifiers of the respondents, we get a new dataset containing 12245 observations and 273 variables. Generating fully synthetic data for each of our datasets using the *syn()* function in its mostly default parameter settings returns the four fully synthetic datasets of the same dimensions as their original counterparts, all generated using the CART algorithm method. Table 1 gives an overview of the relevant variables used throughout the data analysis regarding the evaluation of the utility and assessing the disclosure risk for the generated synthetic datasets.

Name	Label	Range of Values
adchb	Additional child benefit	[0; 19596]
alg2	Unemployment benefit II	[0; 56531]
asyl	Asylum seeker benefit	[0; 39600]
chspt	Child allowance	[0; 35328]
d11101	Age of Individual	[0; 105]
d11102ll	Gender of Individual	1 = Male, 2 = Female
d11104	Marital Status of Individual	1 = Married, 2 = Single, 3 = Widowed, 4 = Divorced, 5 = Separated
d11106	Number of Persons in HH	[1; 17]
d11107	Number of Children in HH	[0; 11]
d11109	Number of Years of Education	[7; 18]
divdy	Interest, dividend income	[0; 10741452]
e11101	Annual Work Hours of Individual	[0; 8314]
e11105_v1	Occupation of Individual (ISCO88)	[0; 9330]
hgacquis	Means Of Acquiring Dwelling	[1; 4]
hghinc	Monthly Household Net Income (EUR)	[0; 1700000]
hgmoveyr	Year Moved Into Dwelling	[1900; 2021]
hgowner	Tenant Or Owner Of Dwelling	[1; 5]
hgrent	Amount Of Rent Minus Heating Costs	[0; 7318]
hgsize	Size Of Housing Unit In Sq M	[6; 938]
i11101	HH Pre-Government Income	[0; 13345989]
i11105	HH Imputed Rent	[0; 87211.0]
i11109	Total HH Taxes	[0; 4719104]
l11101	State of Residence	[0; 16]
m11102	Inpatient nights in hospital	[0; 365]
m11124	Disability Status of Individual	[0; 1]
m11125	Satisfaction with Health	[0; 10]

m11127	Number of annual doctor visits	[0; 484]
p11101	Overall life satisfaction	[0; 10]
pgausb	Required Training For Job	[0; 9]
pgbilzeit	Amount Of Education Or Training In Years	[7; 18]
pgemplst	Employment Status	1 = Full-Time Employment, 2 = Regular Part-Time Employment 3 = Vocational Training, 4 = Marginal, Irregular Part-Time Employment, 5 = Not Employed
pgerwzeit	Length Of Time With Firm	[0; 67.75]
pgfamstd	Marital Status In Survey Year	[1; 8]
pgisco88	Current Occupational Classification (ISCO-88 Com)	[0; 9330]
pgjobch	Occupational Change	[1;5]
pgkldb92	Current Occupational Classification (KldB92)	[110; 9971]
pglabgro	Current Gross Labor Income in Euro	[0; 2700000]
pglabnet	Current Net Labor Income in Euro	[0; 1700000]
pgnace	Industry Occupation [pbra] (NACE Rev. 1.1, Sector)	[1; 99]
pgnation	1. Nationality	[1; 999]
pgpsbil	School-Leaving Degree	[1; 8]
pgstib	Occupational Position	[10; 640]
pguebstd	Overtime Per Week	[0; 57]
pgvebzeit	Agreed Upon Work Time Per Week	[0; 80]

Table 1: Overview of the variables used throughout the analyses done on the SOEP datasets with information on their labels and value range (Grabka, 2020; SOEP Group, 2021, 2021).

4.2 Utility Evaluation

The global and outcome utility are analyzed to evaluate how useful the datasets are. The utility metrics pMSE, S_pMSE, and SPECKS are considered for each data set to assess the global utility. The results of three different regression analyses are compared to evaluate the outcome utility.

The first analysis observes the relationships between life satisfaction and factors like marital status, gross labor income, agreed-upon work time per week, actual annual working hours, employment status, and household income.

$$p11101 \sim d11104 + pglabgro + pgvebzeit + e11101 + pgemplst + hghinc$$

The second analysis examines the relationships between health satisfaction and factors such as age, marital status, annual number of doctor visits, and nights spent in the hospital.

$$m11125 \sim d11101 + d11104 + m11127 + m11102$$

The third analysis explores, how marital status, years of education, and gender relate to household income.

$$hghinc \sim d11104 + d11109 + d11102ll$$

4.2.1 Global Utility

In evaluating the global utility of the synthetic datasets across the variables pgen, hgen, and pequiv and their aggregated forms compared to the originals, it is observable in Table 2, that hgen yields the best values among the three utility values, followed by pequiv, the merged data set, and lastly the pgen dataset.

	pMSE	**S_pMSE**	**SPECKS**
pgen	0.081	60.707	0.490
hgen	0.011	1.979	0.170
pequiv	0.016	17.658	0.183
Merged dataset	0.081	33.416	0.502

Table 2: Global utility metrics for the complete generated datasets from the SOEP

Based upon the suggestions of Raab et al. (2021, p. 12), the S_pMSE should be kept below ten and ideally below 3. Considering this, while the hgen dataset has an outstanding value of 1.979, the S_pMSE of the other three datasets are all above the recommended threshold of 10. However, since not all variables of each dataset are used for analytical purposes, an evaluation considering

only variables of importance for the regression analyses of Section 4.2.2 can be seen in Table 3, Table 4, and Table 5.

	pMSE	S_pMSE	SPECKS
p11101	4.892e-05	1.917	0.009
d11104	2.986e-05	1.170	0.005
pglabgro	3.097e-05	1.214	0.010
pgvebzeit	5.228e-05	5.121	0.008
e11101	33.137e-05	21.641	0.032
pgemplst	1.965e-05	0.962	0.005
hghinc	12.215e-05	4.786	0.019

Table 3: One-way comparison of variables used in the outcome utility analysis of life satisfaction

When examining global utility by making a one-way comparison of only the variables used in the regression analysis for life satisfaction (see Table 3), only the synthetic variable e11101, which holds information about annual hours of work, has a S_pMSE above 10 with a value of 21.641. However, given the small value of 0.032 for the SPECKS metric, it is difficult to evaluate whether the variable might be problematic. The remainder of the variables have S_pMSE values below 10, with four of the remaining six variables having S_pMSE values below 3.

	pMSE	S_pMSE	SPECKS
m11125	1.877e-05	0.919	0.008
d11101	2.627e-05	1.286	0.008
d11104	2.986e-05	1.170	0.005
m11127	1.773e-05	1.158	0.007
m11102	0.317e-05	0.621	0.000

Table 4: One-way comparison of variables used for the health satisfaction outcome utility.

Table 4 shows the results of a one-way comparison of the variables used in the regression analysis health satisfaction. With all the S_pMSE values of the respective variables being under 3, the global utility of these variables is excellent.

	pMSE	S_pMSE	SPECKS
hghinc	12.214e-05	4.786	0.019
d11104	2.986e-05	1.170	0.005
d11109	4.611e-05	2.258	0.011
d11102ll	0.131e-05	0.256	0.002

Table 5: One-way comparison of variables used for the outcome utility analysis of household income

Table 5 shows the utility metrics of the one-way comparisons for the variables used in the household income regression analysis. Since all variables have S_pMSE values below ten and low Specks values, the global utility for these variables can also be stated as good.

	pMSE	S_pMSE	SPECKS
Life Satisfaction	0.003	4.127	0.074
Health Satisfaction	0.001	1.388	0.043
Household income	0.002	3.855	0.067

Table 6: Global utility metrics for the reduced generated data sets from the SOEP, considering only relevant variables for the respective analysis

According to the overall global utility for each of the three analyses with their respective variables (see Table 6), the S_pMSE values are all below 10 with small SPECKS values, indicating a good global utility for the datasets reduced to only the essential variables of the analyses.

4.2.2 Outcome Utility

Three linear regression models are analyzed using synthetic and original datasets to evaluate the outcome utility. Simulating a synthetic version of the original data aims to reproduce all its features as closely as possible, including any missing values. This process involves every value in the dataset. Since the missing values were not imputed before synthesis, the pattern of the missing data is synthesized instead (Nowok, 2016; Nowok et al., 2016). To allow for valid statistical inferences with overall more accurate estimators and confidence intervals, the missing data for both the original and synthetic datasets is multiply imputed using the *mice()* function and the CART algorithm method. While theoretically, it is always better to use a higher m as the number of imputations, this comes with higher computational labor and storage demands. According to Buuren (2018, chapter 2.8), creating five to 20 datasets with imputed values for missing data is sufficient for analysis, with the underlying conclusions unlikely to change due to increasing m

beyond even 5. Raghunathan et al. (2017) suggests performing the imputation process 10 times for each missing value. The reasoning behind this is that once 10 iterations have been completed, the imputations should have stabilized to a point where the sequence, in which the variables were imputed, no longer holds much significance (Stuart et al., 2009). In other words, the imputed data should be consistent and reliable after 10 iterations. Given the five imputations, which consist of the imputed values after the ten iterations combined with the observed data, a total of 5×10 iterations are performed in total for each dataset.

Since two different kinds of synthetic data are reviewed, the synthetic dataset created by first generating individual synthetic datasets for pgen, hgen, and pequiv and then aggregating them into a combined synthetic dataset is denoted as synthetic data type 1 for the remainder of the thesis. The second synthetic dataset, created by first aggregating the pgen, hgen, and pequiv datasets and then synthesizing all the data, is denoted as synthetic data type 2.

For the first analysis regarding life satisfaction, the results from the regression analysis can be seen in Table 7 and Table 8.

	Coefficient Original data	**Coefficient Synthetic data 1**	**|ΔCoefficients|**	**CI Overlap**
Intercept	7.732***	7.340***	0.392	0.392
d11104.2	-0.289***	-0.245***	0.044	0.777
d11104.3	-0.248**	-0.149*	0.099	0.681
d11104.4	-0.975***	-1.101***	0.126	0.734
d11104.5	-0.969***	-0.638***	0.331	0.408
pglabgro	0.000***	0.000	0.000	0.206
pgvebzeit	-0.004	-0.001	0.003	0.814
e11101	0.000*	0.000***	0.000	-1.821
pgemplst.2	-0.109	0.001	0.110	0.738
pgemplst.3	0.343**	-0.018	0.361	0.042
pgemplst.4	-0.463**	0.014	0.477	0.194
pgemplst.5	-0.610***	0.000	0.610	-0.433
hghinc	0.000***	0.000	0.000	-0.698

Table 7: Results from the regression analysis of life satisfaction if the datasets are first synthesized and then merged for analysis
Note: *** significant at 0.1% level of significance; ** significant at 1% level of significance; * significant at 5% level of significance

According to the inference results for the original data and the synthetic data type 1 in Table 7, the original and synthetic data estimates are close to each other, except for the variables d11104.4, d11105.5, pgemplst.2, pgemplst.3, pgemplst.4 and pgemplst.5, in which the difference in the coefficient estimates is langer than 0.1 (0.126, 0.331, 0.110, 0.361, 0.477, and 0.610, respectively). The significance level of the original dataset is the same as the synthetic dataset for the following variables: pgvebzeit, the agreed -upon work time; pgemplst.2, the dummy variable that indicates the employment status of regular part -time employment; and most of the dummy variables of marital status d11104, except for d11104.3, which stands for a widowed marital status. All the other variables differ in their significance level when using the synthetic dataset. While the CI overlap for the estimates of pgvebzeit is relatively high, with its value of 0.814 being over 0.8, seven out of 12 variable estimates fall below a CI overlap of 0.5, with three out of the seven variables showing no overlap at all.

	Coefficient Original data	**Coefficient Synthetic data 2**	**|ΔCoefficients|**	**CI Overlap**
Intercept	7.732***	7.59***	0.142	0.799
d11104.2	-0.289***	-0.31***	0.021	0.894
d11104.3	-0.248**	-0.385***	0.140	0.602
d11104.4	-0.975***	-0.681***	0.294	0.406
d11104.5	-0.969***	-0.674***	0.295	0.459
pglabgro	0.000***	0.000**	0.000	0.768
pgvebzeit	-0.004	0.001	0.005	0.693
e11101	0.000*	0.000***	0.000	0.725
pgemplst.2	-0.109	-0.128	0.019	0.954
pgemplst.3	0.343**	0.340**	0.003	0.989
pgemplst.4	-0.463**	-0.336*	0.127	0.803
pgemplst.5	-0.610***	-0.384***	0.226	0.530
hghinc	0.000***	0.000***	0.000	0.586

Table 8: Results from the regression analysis of the life satisfaction if the data sets are merged and then synthesized on the complete data
Note: *** significant at 0.1% level of significance; ** significant at 1% level of significance; * significant at 5% level of significance

Table 8 shows the inferential analysis results for the original data and synthetic data type 2. The difference in estimates between the original and synthetic data is small, with six out of 12 variables having a difference above 0.1. Apart from pglabro, e11101, and pgemplst4, the significance levels of the variables remain at the same level for the synthetic data compared to the original. The CI overlaps for the estimates are quite high, with d11104.2, pgemplst.2, pgemplst.3, and pgemplst.4 having values above 0.8, and only d11104.4 and d11104.5 having values below 0.5.

As for the second regression analysis concerning the health satisfaction of the respondents, Table 9 and Table 10 show the results of the estimates and CI overlaps for the original and synthetic datasets.

	Coefficient Original data	**Coefficient Synthetic data 1**	**|ΔCoefficients|**	**CI Overlap**
Intercept	9.549***	9.208***	0.341	-0.495
d11101	-0.049***	-0.043***	0.006	-0.232
d11104.2	-0.023	-0.200***	0.177	0.116
d11104.3	0.139	-0.156*	0.295	0.079
d11104.4	-0.258	-0.066	0.192	0.591
d11104.5	-0.117	-0.421**	0.244	0.459
m11127	-0.032***	-0.025***	0.007	-1.679
m11102	-0.025***	-0.024***	0.001	0.778

Table 9: Results from the regression analysis of health satisfaction if the datasets are first synthesized and then merged for analysis
Note: *** significant at 0.1% level of significance; ** significant at 1% level of significance; * significant at 5% level of significance

Table 9 represents the analysis results for the original data and synthetic data type 1. Only the variables for the age of the individual (d11101), the number of annual doctor visits (m11127), and nights spent in the hospital (m11102) have very close estimates for the original and synthetic dataset, with the difference in coefficient estimates being below 0.01, while upholding the same levels of significance in both datasets for each of these three variables. The dummy variables for marital status d11104 all show differences in estimates above 0.1 and different significance levels

for the original and synthetic datasets, except for the dummy variable for the marital status being divorced d11104.4. The CI overlaps for the estimates are small, with six out of seven variables showing values below 0.5. D11101 and m11127 show no CI overlap at all, with negative values of -0.232 and -1.679, respectively.

	Coefficient Original data	**Coefficient Synthetic data 2**	**|ΔCoefficients|**	**CI Overlap**
Intercept	9.549***	9.309***	0.240	0.158
d11101	-0.049***	-0.044***	0.005	0.247
d11104.2	-0.023	0.027	0.050	0.786
d11104.3	0.139	-0.109	0.248	0.354
d11104.4	-0.258	-0.064	0.194	0.625
d11104.5	-0.117	-0.086	0.091	0.948
m11127	-0.032***	-0.030***	0.002	0.509
m11102	-0.025***	-0.020***	0.005	0.283

Table 10: Results from the regression analysis of health satisfaction if the datasets are merged and then synthesized on the complete data
Note: *** significant at 0.1% level of significance; ** significant at 1% level of significance; * significant at 5% level of significance

Table 10 shows the same analysis results using the same variables as Table 9, with the difference that the original data is compared to synthetic data type 2. The significance levels for each variable are the same for the original and synthetic datasets. Minor differences can be observed in the estimates for d11101, m11127, and m11102, with each having a difference of under 0.01. It can also be seen, that only the two dummy variables, d11104.4 for marital status of divorce and d11104.5 for marital status of being separated, show an estimated difference of more than 0.1. The dummy variable d11104.5 has a high CI overlap of 0.948; it is the only variable with a value above 0.8. Only three out of the seven variables have a CI overlap below 0.5.

The results for the third analysis regarding the household income regression analysis can be seen in Table 11 and Table 12.

	Coefficient Original data	**Coefficient Synthetic data 1**	**|ΔCoefficients|**	**CI Overlap**
Intercept	822.138***	1320.116***	498.028	-2.007

d11104.2	-32.645	6.922	39.567	0.574
d11104.3	-457.856***	-54.185	403.671	-1.634
d11104.4	-346.937***	-2.885	344.052	-0.616
d11104.5	-300.546***	-21.041	279.505	-0.028
d11109	71.320***	5.864	65,456	-3.402
d11102ll.2	-34.578	17.332	51.901	0.245

Table 11: Results from the regression analysis of household income if the datasets are first synthesized and then merged for analysis
Note: *** significant at 0.1% level of significance; ** significant at 1% level of significance; * significant at 5% level of significance

Table 11 shows the regression analysis results for the original data and synthetic data of type 1. The only significance levels preserved from the original data analysis are of the dummy variables of marital status being single (d11104.2) and gender of the individual being female (d11102ll.2), as neither are significant in the synthetic data either. D11104.3, d11104.4, d11104.5, and d11109 all lose their strong significance levels in the synthetic data while having a very high difference in estimates, which is in the hundreds of euros for the three dummy variables for marital status. The CI overlap of the estimates could be better, as five out of six variables show a CI overlap below 0.5, while four of these five variables have no overlap in their estimates.

	Coefficient Original data	**Coefficient Synthetic data 2**	**|ΔCoefficients|**	**CI Overlap**
Intercept	822.138***	996.661***	144.523	0.171
d11104.2	-32.645	-49.130	16.485	0.884
d11104.3	-457.856***	-417.898***	39.958	0.780
d11104.4	-346.937***	-248.593***	98.344	0.600
d11104.5	-300.546***	-233.185***	67.361	0.778
d11109	71.320***	55.928***	15.392	0.175
d11102ll.2	-34.578	-62.077**	27.499	0.671

Table 12: Results from the regression analysis of household income if the datasets are first merged and then synthesized on the complete data
Note: *** significant at 0.1% level of significance; ** significant at 1% level of significance; * significant at 5% level of significance

Table 11 shows the regression analysis results for the original data and synthetic data type 2. Contrary to the results in Table 10, the significance level is the same for the variables in the synthetic data, except for the dummy variable d11102ll.2. The difference in estimates is far smaller as well, with not a single difference in the variables being over a 100. The CI overlap of the estimates is relatively high, with all values being above 0.5, except for number of years of education (d11109), where the overlap is only 0.175. The CI overlap for the dummy variable for marital status single is 0.884, which is higher than 0.8.

4.3 Risk Assessment

Since fully synthetic data is generated from models, there is no direct link or mapping back to the original data. Furthermore, there is no need for the synthetic dataset to be the same size as the original dataset (Drechsler and Haensch, 2023). Because of this lack of a direct link, the usual methods for measuring re-identification risk, which is the likelihood of tracing back data to a specific individual, do not apply to fully synthetic data. However, while fully synthetic data does offer plenty of privacy protection, it is not immune to privacy risks (Drechsler and Haensch, 2023). Understanding and measuring these risks is a challenging task and an area of ongoing research.

4.3.1 Checking for duplicated records

Manrique-Vallier and Hu (2018) show that the FCS approach using a synthesis based on the CART method might duplicate the majority of the original data's records. This problem occurs when the approach always conditions on all other variables in the dataset, according to Drechsler and Haensch (2023). Consequently, it could result in synthetic datasets that are too similar to the original.

When checking for duplicates in the datasets, it was found that while the hgen, pequiv, and aggregated datasets show no replicated records in the synthetic dataset, there is one duplicated record in the pgen dataset that has been fully replicated in the synthetic dataset for the individual with the personal ID 405301. Although the synthetic pgen data has four more records sharing the same specific personal ID, the specific combination of attributes makes the record unique in both the original and synthetic dataset. According to Drechsler and Haensch (2023), if respondents were to find a synthetic record that precisely mirrored their own, they might feel that their privacy had been compromised, even if the match was coincidental. This perceived risk could undermine trust in the data collection process and potentially discourage individuals from participating in future surveys.

4.3.2 K-Anonymity

To determine how well disclosure risk is upheld in the definition of k-anonymity, a combination of key variables is assigned to each dataset. The following are defined as key variables for the datasets (further information for each variable is in Table1):

- pgen: pgfamstd, pgstib, pgemplst, pgjobch, pgnace, pgisco88, pgkldb92, pgausb, pgpsbil, pglabnet, pgerwzeit, pguebstd, pgbilzeit
- hgen: hgacquis, hgowner, hgmoveyr, hgrent, hgsize
- pequiv: d11102ll, d11104, e11105_v1, l11101, d11101, d11106, d11107, d11109, e11101, adchb, alg2, asyl, chspt, divdy, i11105, i11109

As for the merged datasets, all the abovementioned variables are selected as key variables.

	Original data set			**Synthetic data set**		
	2-anonymity	3-anonymity	5-anonymity	2-anonymity	3-anonymity	5-anonymity
pgen	4055 (33.116%)	5378 (43.920%)	6314 (51.564%)	4512 (36.848%)	5807 (47.423%)	6680 (54.553%)
hgen	0 (0.000%)	0 (0.000%)	0 (0.000%)	0 (0.000%)	0 (0.000%)	0 (0.000%)
pequiv	0 (0.000%)	0 (0.000%)	0 (0.000%)	0 (0.000%)	0 (0.000%)	0 (0.000%)
merged data (syn type 1)	7023 (57.354%)	7825 (63.904%)	8472 (69.187%)	6569 (17.027%)	13705 (35.525%)	21241 (55.058%)
merged data (syn type 2)	7023 (57.354%)	7825 (63.904%)	8472 (69.187%)	7515 (61.372%)	8084 (66.019%)	8643 (70.584%)

Table 13: Absolute and relative numbers of observations that violate the anonymity with k=2, 3, or 5 for each dataset

The combined key variables for the hgen and pequiv dataset fulfill 5-anonymity, meaning all individuals' key variables are shared with at least four others. As for the key variables for the pgen dataset, 33.116% of the records violated the 2-anonymity, 43.920% the 3-anonymity, and 51.564% the 5-anonymity in the original dataset. The percentage rises by up to 5% for each anonymity level for the synthetic version of the pgen dataset. A significant portion of the original merged dataset violates the 2-anonymity, 3-anonymity, and 5-anonymity conditions, meaning that many records have unique or nearly unique combinations of key variables. This could potentially be a significant

privacy concern if the original dataset is left untouched. While the synthetic dataset type 2 also has a similar number of records violating the 2-anonymity, 3-anonymity, and 5-anonymity conditions, the synthetic dataset type 1 vastly reduces the number of records violating the anonymity conditions, with the number of 3-anonymity violations being almost half that in the original and the 2-anonymity violations being less than a third of that in the original data.

The synthetic datasets still contain unique combinations of key variables since the method is not built around upholding the k-anonymity definition. However, these combinations still pose a shallow risk of disclosure since these values are not traceable to the original respondent as they are fully synthetic.

4.3.3 L-Diversity

Adding to the definition of k-anonymity, the distribution of sensitive variables within a group of observations sharing the same pattern in key variables is examined as l-diversity (Templ, 2017). When observing the same combination of key variables for the datasets as in Section 4.3.2, the following sensitive variables are reviewed for the respective datasets:

- pgen: pgnation
- hgen: hghinc
- pequiv: m11102, m11124, i11101
- merged datasets: all five variables mentioned above

	Sensitive variables	**Original dataset**				**Synthetic dataset**			
		Min.	Median	Mean	Max.	Min.	Median	Mean	Max.
pgen	pgnation	1.000	1.000	1.023	4.000	1.000	1.000	1.014	3.000
hgen	hghinc	1.000	1.000	70.310	262.000	1.000	1.000	60.100	228.000
pequiv	m11102	1.000	1.000	5.353	34.000	1.000	1.000	4.847	28.000
	m11124	1.000	1.000	1.608	6.000	1.000	1.000	1.585	6.000
	i11101	1.000	2.000	44.100	452.000	1.000	2.000	40.940	386.000
merged data (synthetic	pgnation	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	hghinc	1.000	1.000	1.004	3.000	1.000	1.000	1.039	6.000
	m11102	1.000	1.000	1.001	2.000	1.000	1.000	1.004	3.000

data type 1)	m11124	1.000	1.000	1.000	1.000	1.000	1.000	1.002	2.000
	i11101	1.000	1.000	1.004	3.000	1.000	1.000	1.014	5.000
merged data (synthetic data type 2)	pgnation	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	hghinc	1.000	1.000	1.004	3.000	1.000	1.000	1.002	3.000
	m11102	1.000	1.000	1.001	2.000	1.000	1.000	1.000	1.000
	m11124	1.000	1.000	1.000	1.000	1.000	1.000	1.001	2.000
	i11101	1.000	1.000	1.004	3.000	1.000	1.000	1.002	3.000

Table 14: L-diversity measures showing the minimum, median, mean, and maximum values of l-diversity

Table 14 shows basic statistical measures of the distinct l-diversity for the respective sensitive variables. The values for the measures of the original and synthetic data measures are very similar. The minimum and median values of l-diversity in the data are 1 for all the original and synthetic variables, except for i11101 in the pequiv data, for which it is 2. This indicates that in at least 50% of the groups, all records have the same sensitive attribute, aside from i11101. Furthermore, all records have the same sensitive detail for at least one group of records indistinguishable by their key variables. The mean values are primarily distributed around approximately one, save for the three sensitive variables hghinc, m11102, and i11101 in the pequiv dataset. These three variables have the highest maximum value for l-diversity as well. The highest difference for the l-diversity between the original and synthetic datasets can be observed in the diversity values for i11101, for which the maximum values show a difference of 66, and the diversity values for hghinc, for which the mean value shows a difference of approximately 10.

5 Conclusion

Given the growing importance of data privacy for statistical institutions in modern times, it has become essential to consider effective and practical ways of anonymizing microdata that contains detailed information about individuals or organizations, while ensuring its utility for research purposes. This thesis focused on SDC as an essential field of privacy protection for sensitive data. The method explored was the generation of fully synthetic data, which aims to simulate real data without risking the exposure of the real identities of the respondents. Using three datasets containing information from personal, household, and health questionnaires from the SOEP, an evaluation of the utility and risk of disclosure in the synthetic data was performed while also doing a quality comparison of synthetic data when datasets are synthesized before and after merging.

The utility evaluation considered both global utility and outcome utility, comparing synthetic data to the original both in a general sense and in relation to specific analyses. As for global utility, the utility metric values over all variables of each dataset were subpar in quality for each synthetic dataset, save for the hgen dataset. However, when analyzing utility again for a reduced number of variables containing only relevant predictors and outcome variables for the regression analyses later performed for the outcome utility, the metrics for the S_pMSE and SPECKS were very good, indicating similar distributional patterns between the synthetic and original datasets for these variables. For the outcome utility, linear regression analyses for life satisfaction, health satisfaction, and household income were evaluated for original and synthetic data. It is apparent, that synthesizing the individual datasets to aggregate them into one synthetic dataset yielded inferential results that are far worse than the results obtained on the dataset created in the reverse order. The CI overlaps for the estimates were often small to nonexistent, with the variables often having different significance levels in the original and synthetic datasets, leading to different conclusions in the eyes of a researcher. Compared to that, inferential results gained on data obtained by first aggregating individual datasets and then synthesizing the aggregated dataset were better, as the estimates yielded higher CI overlaps and similar significance levels between the original and synthetic data.

Disclosure risk was assessed using the principles of k-anonymity and l-diversity to gauge how well the synthetic data maintained confidentiality and privacy. Additionally, the perceived risk of respondents was compared between the original and synthetic data. Since synthetic data generated by the FCS approach using the CART synthesizer might lead to more frequent copying of records from the original data into the synthetic data, an evaluation of how many records of the original

data were duplicated into the synthetic data was performed. Apart from the pgen dataset, in which one record was found to have a unique match in the original data, the synthetic datasets did not have any further unique matches in the original data. As for k-anonymity, the number of observations, that violate k-anonymity for k = 2, 3, and 5 was very similar between the original and synthetic data. The only significant difference could be seen in the number of observations with k-anonymity violations, with a drastically lowered number of violations for the 2- and 3-anonymity given the synthetic data, in which the individual datasets were first synthesized and then merged into one. Considering the significant portion of records with unique combinations of key variables in the original datasets except for the hgen and pequiv dataset, a potential privacy concern could arise if left untouched. While the generation of synthetic data itself does not significantly improve k-anonymity, it still mitigates the disclosure risk significantly since the unique combinations of key variables are synthetic values, that break the linkage to the original respondents. Looking at the l-diversity measures of the sensitive variables for the original and synthetic data, no significant improvement was observed for the synthetic datasets. Thus, approximately the same magnitude of diversity exists in sensitive attributes for synthetic and original data within a group of records, that are identical to each other according to certain key variables.

Considering all the aspects mentioned, based on the results in this thesis, fully synthetic data provides acceptable utility on the condition, that relevant data is first polished and linked and then a synthesizer is applied to the dataset as a whole. The re-identification risk remains low even with unique combinations of key identifiers due to breaking the linkage to original respondents with the simulated values. However, protecting sensitive variables falls short as there is no vast improvement in the l-diversity sense.

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